

PART 1 - EVALUATION OF QUIZ GENERATION STRATEGIES

1. OBJECTIVE

The objective of Part 1 was to determine which of the three quiz-generation strategies produces the most useful and reliable set of questions from the provided lecture material.

Rather than selecting a strategy based on a small number of questions that appeared better subjectively, I evaluated the generated questions systematically. The evaluation was performed at two levels:

1. Individual question quality — whether each question was grounded, correct, clear, appropriately challenging, and cognitively useful.
2. Question-set quality — whether the complete set provided sufficient topic coverage, avoided redundancy, contained a useful range of cognitive levels, and produced a sufficiently high proportion of usable questions.

The three strategies provided were:

- Strategy A: The complete lecture was given to the model and questions were generated directly.
- Strategy B: Relevant passages were retrieved first and questions were generated using those passages.
- Strategy C: Questions were generated first and then passed through a second verification step that removed questions that could not be supported by the source.

The supplied question sets contained 12 questions from Strategy A, 12 questions from Strategy B, and 8 questions from Strategy C.

2. EVALUATION METHODOLOGY

I first created a question-level scoring rubric consisting of five criteria:

- Groundedness
- Correctness
- Clarity
- Cognitive Level
- Difficulty

Each criterion was scored on a 0–2 scale, giving every question a maximum possible score of 10.

Scoring rubric:

Groundedness:

- 0 = Unsupported or contradictory
- 1 = Partially supported or requires additional inference
- 2 = Clearly supported by the lecture

Correctness:

- 0 = Incorrect
- 1 = Minor issue or imprecision
- 2 = Correct

Clarity:

- 0 = Ambiguous or confusing
- 1 = Understandable but could be improved
- 2 = Clear and unambiguous

Cognitive Level:

- 0 = Recall
- 1 = Understanding
- 2 = Application / Reasoning

Difficulty:

- 0 = Poorly calibrated
- 1 = Reasonably calibrated
- 2 = Appropriately challenging

Groundedness was evaluated specifically against the supplied lecture rather than against general domain knowledge. This was important because a question can be factually true in the real world while still being inappropriate if the answer cannot be obtained from the professor's lecture.

Difficulty was treated as an expert judgment because there was no student response data available to empirically measure question difficulty.

After scoring the individual questions, I performed additional analysis at the strategy level using:

- Topic Coverage
- Redundancy
- Cognitive Diversity
- Usable Question Yield
- Critical Failure Rate

These metrics were then normalized to a common 0–10 scale and combined using a weighted scoring scheme.

3. QUESTION-LEVEL QUALITY

The average question-quality scores were:

Strategy A: 6.67 / 10

Strategy B: 8.17 / 10

Strategy C: 9.25 / 10

Strategy C produced the highest-quality questions on average.

Strategy B also performed relatively well at the individual-question level, while Strategy A had the lowest average score.

One of the main reasons for Strategy A's lower score was the presence of questions that sounded plausible but were not actually supported by the supplied lecture.

Examples include:

- A04 made a stronger claim about Adam being the best choice for deep learning tasks than the lecture supported.
- A07 introduced a specific batch size of 256 that was not provided in the lecture.
- A11 introduced a publication year that was not contained in the lecture.

These examples demonstrate why groundedness was treated as an independent criterion. A question does not become suitable for a professor's quiz simply because its wording sounds technically reasonable.

4. TOPIC COVERAGE

After scoring individual questions, I mapped the questions to the major concepts represented in the lecture.

The analysis considered concepts such as:

- Gradient descent
- Learning rate
- Batch and mini-batch gradient descent
- Backpropagation
- Chain rule
- Vanishing gradients
- Sigmoid

- ReLU
- Momentum
- Adam
- Training diagnostics

The resulting topic coverage was:

Strategy A: 100%

Strategy B: 20%

Strategy C: 80%

Strategy A provided the broadest coverage of the lecture.

However, broad coverage came with a trade-off. Strategy A also contained several unsupported questions, meaning that covering more topics does not automatically make the resulting quiz more reliable.

Strategy B was much more concentrated around the sigmoid derivative, vanishing-gradient, and ReLU portions of the lecture. This meant that many of its individual questions were strong, but the set did not represent the lecture broadly.

Strategy C covered most of the identified concepts while also producing stronger individual questions.

This demonstrates an important distinction: individual question quality and overall question-set coverage are separate properties.

5. REDUNDANCY ANALYSIS

I next evaluated whether questions within each strategy were substantially testing the same underlying concept.

I did not classify questions as redundant merely because they belonged to the same broad topic. Instead, redundancy was identified when multiple questions were substantially testing the same knowledge or reasoning.

The results were:

Strategy A:

0 redundant questions out of 12

Redundancy rate = 0%

Strategy B:

4 redundant questions out of 12
Redundancy rate = 33.3%

Strategy C:
0 redundant questions out of 8
Redundancy rate = 0%

Strategy B showed the clearest redundancy problem.

Several B questions repeatedly focused on closely related aspects of the sigmoid derivative and vanishing-gradient mechanism. Although these questions were not necessarily incorrect, they reduced the amount of distinct knowledge being tested by the set.

Strategies A and C had no observed redundancy under the definition used in this evaluation.

This metric was important because a question bank with many questions is not necessarily more useful if several questions are effectively testing the same thing.

6. COGNITIVE DIVERSITY

I also examined the cognitive level of the generated questions.

Questions were classified into:

- Recall
- Understanding
- Application / Reasoning

The distribution was:

Strategy A:
Recall = 8
Understanding = 3
Application / Reasoning = 1

Strategy B:
Recall = 6
Understanding = 6
Application / Reasoning = 0

Strategy C:
Recall = 0
Understanding = 6

Application / Reasoning = 2

Strategy A was strongly dominated by recall questions.

Strategy B had a more balanced distribution between recall and understanding, but did not produce questions classified as application or reasoning.

Strategy C produced a much more conceptual set, with most questions classified as understanding and two classified as application/reasoning.

I did not assume that a higher cognitive level is automatically better. A good quiz can contain recall questions as well as questions requiring deeper reasoning. Therefore, cognitive diversity was used as a comparative measure rather than as a claim that there is one universally correct distribution.

7. USABLE QUESTION YIELD

Because the strategies produced different numbers of questions, I also evaluated how many of the supplied questions were considered usable after scoring.

Strategy A:

9 usable questions out of 12

Usable yield = 75%

Strategy B:

12 usable questions out of 12

Usable yield = 100%

Strategy C:

8 usable questions out of 8

Usable yield = 100%

Strategy A therefore had the lowest usable-question yield.

B and C had 100% usable yield among the questions that were supplied for evaluation.

However, the result for C needs to be interpreted carefully because only the final retained questions were supplied to us. We do not know how many questions C generated before its verification step or how many it discarded.

Therefore, C's 100% usable yield refers specifically to the eight questions we received, not to everything C originally generated.

8. CRITICAL FAILURE RATE

I also tracked questions with critical grounding or correctness problems.

Strategy A:

3 critical failures out of 12

Critical failure rate = 25%

Strategy B:

0 critical failures out of 12

Critical failure rate = 0%

Strategy C:

0 critical failures out of 8

Critical failure rate = 0%

Strategy A had three critical failures, corresponding to a 25% critical failure rate.

B and C had no observed critical failures in their supplied final question sets.

This was particularly important because unsupported questions are more problematic than merely mediocre questions. If a generated question introduces information that is not present in the professor's material, it can undermine trust in the entire generated quiz.

9. FINAL WEIGHTED COMPARISON

To combine the different evaluation dimensions, I converted the relevant metrics to a common 0–10 scale and applied the following weights:

Mean Question Quality = 40%

Topic Coverage = 20%

Redundancy = 15%

Cognitive Diversity = 10%

Critical Failure = 15%

Total = 100%

The resulting scores were:

Strategy A = 8.29 / 10

Strategy B = 6.87 / 10

Strategy C = 9.10 / 10

For Strategy A:

$$\begin{aligned} & (6.67 \times 0.40) \\ & + (10.00 \times 0.20) \\ & + (10.00 \times 0.15) \\ & + (10.00 \times 0.10) \\ & + (7.50 \times 0.15) \end{aligned}$$

$$= 8.29 / 10$$

For Strategy B:

$$\begin{aligned} & (8.17 \times 0.40) \\ & + (2.00 \times 0.20) \\ & + (6.67 \times 0.15) \\ & + (7.00 \times 0.10) \\ & + (10.00 \times 0.15) \end{aligned}$$

$$= 6.87 / 10$$

For Strategy C:

$$\begin{aligned} & (9.25 \times 0.40) \\ & + (8.00 \times 0.20) \\ & + (10.00 \times 0.15) \\ & + (8.00 \times 0.10) \\ & + (10.00 \times 0.15) \end{aligned}$$

$$= 9.10 / 10$$

Final ranking:

1. Strategy C — 9.10 / 10
2. Strategy A — 8.29 / 10
3. Strategy B — 6.87 / 10

10. OVERALL ASSESSMENT OF STRATEGY A

Strategy A's main strength was its breadth.

It covered essentially the full range of the lecture, giving it the highest topic coverage of the three strategies. It also showed no observed redundancy.

However, its question-level quality was the lowest of the three strategies, and it produced three critical failures.

The main issue with A was therefore not lack of coverage, but reliability. Some questions introduced information that was not actually supported by the lecture.

Strategy A is therefore useful for broad coverage but less attractive as a final generation strategy without an additional quality-control step.

11. OVERALL ASSESSMENT OF STRATEGY B

Strategy B produced relatively strong individual questions and had no observed critical failures.

However, its main weakness was the narrowness of the generated set.

Most of its questions concentrated around a small group of concepts, particularly sigmoid derivatives, vanishing gradients, and ReLU. This resulted in low topic coverage.

It also had the highest observed redundancy rate at 33.3%.

Therefore, B appears capable of producing accurate questions when relevant material is retrieved, but the retrieval-and-generation process, at least for this supplied lecture, did not produce a sufficiently diverse question set.

12. OVERALL ASSESSMENT OF STRATEGY C

Strategy C performed best on the observed questions.

Its main strengths were:

- Highest mean question quality: 9.25/10
- 100% observed usable-question yield
- 0 observed critical failures
- 0 observed redundancy
- Strong cognitive diversity
- 80% topic coverage

The verification step appears particularly useful because it provides an additional opportunity to remove questions that cannot be supported by the source.

However, this comes with the most important limitation in the entire comparison.

13. THE 12 / 12 / 8 PROBLEM

Strategies A and B each provided 12 questions, while Strategy C provided only 8.

This difference cannot simply be ignored.

Strategy C includes a second-pass verification process that removes unsupported questions. We therefore know that some questions were potentially discarded, but the supplied data do not tell us:

- How many questions C generated originally
- Which questions were rejected
- Why each question was rejected
- Whether rejected questions were incorrect, unsupported, redundant, or simply lower quality
- What concepts the rejected questions covered

As a result, we can evaluate the quality of C's observed retained questions, but we cannot completely evaluate the efficiency or overall behavior of the C pipeline.

For example, its observed usable-question yield is 100%, but that does not mean that 100% of all questions generated by C were usable.

This is the main reason I would describe the recommendation as moderate confidence rather than high confidence.

14. RECOMMENDATION

Based on the evidence available, I would choose Strategy C as the strongest candidate to ship.

C produced the strongest observed question set and avoided the critical grounding failures seen in A. It also produced more conceptually demanding questions and showed no observed redundancy.

However, I would not describe C as definitively proven to be the best end-to-end strategy.

The evidence supports a more precise conclusion:

"Strategy C produces the strongest observed final question set, but the available data are insufficient to determine whether it is the best end-to-end pipeline because its rejected output is not provided."

This distinction is important because the assignment asks not only for a recommendation but also for an assessment of how much confidence that recommendation deserves.

15. WHAT WOULD CHANGE MY MIND?

The most useful next experiment would be to evaluate all three strategies under a more controlled setup.

I would:

1. Give all three strategies the same lecture material.
2. Set the same target number of generated questions for each.
3. Record every question initially generated.
4. For Strategy C, record every question rejected during verification and the reason for rejection.
5. Evaluate the final retained questions using the same rubric.
6. Compare both final quality and usable-question yield.

This would allow us to distinguish between two possibilities.

Possibility 1: C genuinely produces better questions.

Possibility 2: C produces many mediocre or unusable questions but its verification stage removes them effectively.

Both outcomes could result in a high-quality final set, but they would imply different conclusions about the underlying generation strategy.

I would therefore want the next experiment to evaluate not only the final output, but also the generation-to-retention pipeline.

16. FINAL CONCLUSION

The evidence from the supplied question sets favors Strategy C.

C achieved the highest overall weighted score of 9.10/10, compared with 8.29/10 for A and 6.87/10 for B.

Its strongest advantages were high question-level quality, strong cognitive diversity, no observed critical failures, and no observed redundancy.

Strategy A's strongest advantage was broad topic coverage, but this was accompanied by a 25% critical failure rate.

Strategy B produced reasonably strong individual questions but suffered from narrow topic coverage and substantial redundancy.

Therefore, if a decision has to be made using the current evidence, Strategy C is the strategy I would ship.

However, I would treat this as a provisional recommendation with moderate confidence. The unequal output sizes and the absence of C's rejected questions prevent a complete comparison of the three generation pipelines.

The next evaluation should therefore focus on the missing evidence: the complete generated output, rejection reasons, retention rate, and performance across multiple lectures.

The key conclusion from this analysis is that question quality alone is not enough. A useful generation strategy needs to produce questions that are grounded, varied, appropriately challenging, and representative of the source material, while also producing enough usable questions to support a real quiz workflow.

The final recommendation is:

SHIP STRATEGY C PROVISIONALLY, BUT RUN A CONTROLLED FOLLOW-UP EVALUATION BEFORE TREATING IT AS THE DEFINITELY BEST GENERATION STRATEGY.